Eduardo Alejandro Aviles Jimenez

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Machine Learning Engineer with 10+ years of experience spanning full-stack development, data engineering, and AI/ML solutions across financial services, social media, and enterprise applications. Expert in building end-to-end ML pipelines, deep learning, NLP, computer vision, and predictive analytics. Proven leader in architecting scalable cloud infrastructure and DevOps practices that reduce operational costs by 30%. Strong background in technical innovation and business impact, delivering production-ready AI solutions with regulatory compliance.

Education

Computer Systems Engineering — *Engineering Degree* Jan 2005 - Sept 2011
Universidad De San Carlos De Guatemala

Work Experience

LoanPro — *Machine Learning Engineer* Jun 2022 - Present

- Implemented a Loan Creation Monitor, cluster user behavior, and identify anomalies, improving tenant targeting.
- Developed an anomaly detection system that reduced tenant noise data by 25%, filtering irrelevant events allowing the business to focus on critical insights.
- Created per-tenant smart threshold calculation, considering data distribution, multiple modes, skewness, and kurtosis, optimizing sensitivity for better decision-making.
- Built automated model workflows with continuous training (CT) and model validation to monitor online model performance, preventing model degradation.
- Developed a custom data pipeline for training and inference, using ECS and Fargate for scalability and reliability.
- Implemented a custom LLM AI assistant to summarize and analyze loan data and provide insights and recommendations.
- Developed Risk Radar project using **XGBoost** to predict loan delinquency across 3-month windows, creating comprehensive risk scores that align with IFRS 9 regulatory requirements for Expected Credit Loss (ECL) provisioning.
- Built predictive models calculating probability-weighted risk scores across three delinquency stages (Current, Early/Severe Delinquency, Default), enabling proactive risk management and reducing potential losses by identifying high-risk loans before default.
- Created Risk Radar dashboard with tenant risk leaderboard and early warning alerts, improving portfolio oversight and enabling data-driven decision making for loan servicing teams.

BairesDev - HipTrain — *Senior Software Engineer* Aug 2021 - Jun 2022

- Led infrastructure, architecture, and CI/CD pipelines for back-end and front-end solutions, improving deployment times by 30%.
- Developed scheduling and reservation core system for training classes, improving scheduling by 80%.
- Built custom generic calendar invite logic for scheduling, improving integration and user convenience.
- Implemented a chatbot using Amazon Lex, which automated 40% of customer inquiries and improved user interaction efficiency.
- Led and managed back-end and infrastructure projects, ensuring scalable architecture and seamless integration with existing systems.

LogMeIn — *BI/ML Developer* Apr 2016 - Dec 2021

- Developed infrastructure for data integration and ETL with multiple sources, improving data processing capabilities.
- Built microservices for custom ETL solutions, enhancing scalability and reliability.
- Implemented ML algorithms for product enhancements, leading to 20% increase in prediction accuracy.

SSNAPP (Social Media Gateways) — *Senior Software Developer* Sep 2014 - Apr 2016

- Led transition from monolithic architecture to microservices using Netflix OSS, increasing system scalability by 50% and reducing downtime by 20%.
- Optimized back-end infrastructure and implemented AWS solutions, reducing infrastructure costs by 30%.

Skills

- **Programming Languages:** Python, Java, TypeScript, JavaScript, SQL
- **Web & Backend Frameworks:** Node.js, React, SolidJS, NestJS, Spring Boot, FastAPI, Django, Flask
- **Machine Learning & AI:** XGBoost, Scikit-Learn, TensorFlow, Keras, MLFlow, Pandas, DVC, Weights& Biases, LlamaIndex
- **Cloud & Infrastructure:** AWS (ECS, Fargate, Lambda), Docker, Kubernetes, Terraform, Jenkins, CloudFormation, Ansible
- **Databases & Storage:** PostgreSQL, Redis, RethinkDB, DynamoDB, Redshift
- **Financial & Compliance:** IFRS 9, Expected Credit Loss (ECL), Regulatory Reporting, Risk Modeling
- **Development Tools:** Git, GitHub Actions, RabbitMQ, Netflix OSS, Amazon Lex

Awards and Certificates

Hackathon/LoanPro - Risk Radar — ***First Place*** - Jan 2025

- Risk Radar: Advanced loan delinquency prediction system using XGBoost with IFRS 9 compliance, featuring probability-weighted risk scoring and automated portfolio monitoring dashboard.

Hackathon/LoanPro - Loan Monitoring — ***First Place*** - Apr 2024

- Loan monitoring system that utilized clustering and anomaly detection techniques to improve tenant targeting.

Hackathon/LogMeIn — ***First Place*** - Jul 2018

- Created an LSTM-based neural network to predict customer churn, leveraging advanced recurrent neural network architectures to solve critical business problems related to customer retention. Also voted as developers' choice.

Nano Degree — ***Machine Learning DevOps Engineer*** Jan 2022 - Apr 2022

- Production-ready ML pipelines, automated workflows with **CI/CD**, **FastAPI**, **MLFlow**, **Weights&Biases**, and model performance tracking.

Nano Degree — ***Cloud DevOps Engineer*** Jul 2019 - Aug 2020

- Infrastructure as code (IaC) with **CloudFormation** and **Ansible**, **CI/CD** pipelines with **Jenkins**, and **Kubernetes** microservices.

Nano Degree — ***Artificial Intelligence Nanodegree*** Jan 2018 - Jul 2018

- AI algorithms, search optimization, planning, and pattern recognition.

Nano Degree — ***Deep Learning Nanodegree*** Jul 2017 - Aug 2017

- Image classification, sentiment analysis, and face generation using CNNs, RNNs, and GANs.

Technical Projects

LoanPro — ***Risk Radar - Loan Delinquency Prediction System*** Jan 2025 - Present

- Developed comprehensive loan delinquency prediction system using **XGBoost** models to calculate probability-weighted risk scores across three IFRS 9-compliant stages: Current/Performing, Early/Severe Delinquency, and Default/Charged-off.
- Implemented sophisticated risk scoring algorithm that considers multiple delinquency probabilities over 3-month windows, enabling proactive risk management and early intervention strategies.
- Built Risk Radar dashboard with tenant risk leaderboard, automated alerts, and drill-down capabilities, providing actionable insights for loan servicing teams and portfolio managers.

LoanPro — ***Loan Monitoring System*** Jan 2024 - Dec 2024

- Designed and deployed a system to monitor loan creation, cluster user behavior, and identify anomalies, using **MLFlow**, **Python**, **Pandas**, and **Scikit-learn**. Structured data pipeline for Training and Inference with **ECS** and **Fargate** for scalability and reliability.

Personal Project — ***Predict Customer Churn*** Jan 2024 - Apr 2024

- Python package for customer churn prediction, adhering to PEP8 standards with best practices for modularity, documentation, and testing.

Personal Project — ***ML Pipeline for Rental Price Prediction*** Jan 2024 - Apr 2024

- End-to-end ML pipeline for NYC rental price prediction with automated weekly retraining and deployment.

Personal Project — ***Dynamic Risk Assessment System*** Jan 2024 - Apr 2024

- Automated ML model lifecycle management with data ingestion, scoring, retraining, and deployment.

Speaking Engagements

FaceBook dev Circles, Guatemala — ***Neural Network from scratch*** - March 2019

- Implemented neural network without external ML libraries, showcasing fundamental AI concepts with raw Python.

FaceBook dev Circles, Guatemala — ***RNNs and Sequence to Sequence Model*** - Oct 2018

- Custom RNN-based model to generate fake social media threads, trained on ”Velocidad Maxima” dataset.

Mayan Linguistic Conference, Guatemala — ***Neural Machine Translation*** - Jul 2018

- Text translation evolution and neural models for Mayan dialect translation and language preservation.

Google IO Extended, Guatemala — ***Generative Adversarial Networks (GAN)*** - May 2018

- Keynote on GAN implementation for facial image generation, demonstrating GAN versatility across applications.